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التحديات والاستراتيجيات وتقنيات المواجهة

Part three

Assessing Responsible AI Awareness in Omani Law Enforcement: A Scenario-Based Framework and Training Model for Transnational Crime Contexts

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Abstract

This study addresses the growing need to strengthen responsible AI awareness among personnel in security-relevant roles in the context of transnational crime. As policing agencies increasingly rely on AI technologies for surveillance, crime detection, and data analysis, ethical challenges such as data privacy, algorithmic bias, and accountability have emerged, highlighting the need for structured and practical training approaches. To address this gap, this study adopts a mixed-methods, design-based research approach, collecting data from 40 survey participants and three semi-structured interviews in the Omani context. Findings revealed that data privacy was the weakest awareness domain (37.5% baseline correct response rate), and that structured AI ethics training is largely absent from participants' professional environments. Based on these empirical findings, this study proposes two interconnected contributions: the Ethical AI Awareness and Training Framework (EAATF), comprising six sequential principles: Evaluate, Awareness, Apply, Feedback, Reflect, and Update. and the Data-Driven Ethical AI Awareness and Training Model (DDEATM), a personalized, scenario-based training intervention addressing transnational crime contexts including surveillance bias, data privacy, and cross-border accountability. Initial results indicate an improvement in aggregate AI ethics awareness from 76.3% to 80.6% following scenario-based engagement. This study is particularly relevant to the Sultanate of Oman, where advancing AI integration in policing aligns with Oman Vision 2040, supporting innovation, institutional capacity building, and sustainable security governance.

Keywords: responsible AI, AI ethics awareness, law enforcement training, transnational crime, scenario-based learning, DDEATM, Oman Vision 2040

I. Introduction

In the age of Information and modern technologies, Artificial Intelligence (AI), which is a field of Machine Learning (ML), is playing a pivotal role in transforming individuals' lives toward massive automation. AI must be distinguished from the other kinds of automation, since this technology is capable of learning, adapting, and making decisions (Zhang, 2025). Almost all industries and professional and social domains have witnessed a huge adoption of this technology in their fields, including healthcare, education, agriculture, law enforcement, and policing (Díaz-Rodríguez et al., 2023). The breakthroughs that AI has made in various fields aside, recently, this critical field of computer science has been massively employed in Law Enforcement and Policing, and has proposed several enhancements, such as increased accuracy, speed, and efficiency in areas such as vehicle plate detection using Automated Number Plate Recognition (ANPR), individual identification using Automated Facial Recognition (AFR) technology, and predictive policing (Sorell, 2024; Walczak, 2021; Webb et al., 2024).

The stage and extent of AI adoption in law enforcement in countries vary; nations like the United Kingdom, the US, and Germany are utilizing AI in different areas, including predictive policing to anticipate crimes and their hotspots, facial recognition technologies, and matching DNA in crime scenes with stored DNA profiles, using AI to analyze social networks and open source information to identify possible threats to national security, automated plate detection and etc (Agbadamasi et al., 2025; Lunhol & Torhalo, 2024; Sorell, 2024; Turksen & Kafteranis, n.d.; Webb et al., 2024). Identifying criminals, missing individuals, children, and traffic monitoring, as well as predictive policing, have been made possible in India because of the employment of AI in law enforcement and other sectors (Rani, 2024). Compared to the large implementation of AI in law enforcement in the proposed countries, some of the nations, like Indonesia, Nigeria, and Pakistan, are still in the early and experimental stages of AI implementation, and the law enforcement officials of these countries have positive insights toward the benefits of AI in law enforcement (Kawu, 2025; Soomro et al., 2026; Zuwanda et al., 2024).

Regardless of the degree of AI utilization and its applications in law enforcement in various nations, AI poses some critical challenges that affect individuals' lives. Data and algorithmic biases, privacy issues and violations, inaccurate and unfair decisions and inequality, lack of transparency and accountability that leads to black-box operation, can be mentioned as the challenges of lacking an ethical and responsible AI in law enforcement (Webb et al., 2024; Zhang, 2025; Zuwanda et al., 2024). Although some of the advanced regions, like the European Union and the United States, are showing interest in the use of AI and strategies to govern these technologies through funding research projects, policy initiatives, and existing legal frameworks, there is a significant gap among this attention and the actual achievements of trustworthy AI. In the context of the EU, the majority of the funded research projects related to AI in law enforcement have concentrated on the technological aspect rather than on key ethical principles, like human rights, fairness, transparency, accountability, and public participation (Sanz-Urquijo et al., 2023). As for the United States, the current laws, such as the Federal Trade Commission Act, and the proposed Algorithmic Accountability Act, and sector-specific legislation like HIPAA, are insufficient to address the problems of AI systems (Agbadamasi et al., 2025). Thus, it ought to be mentioned that despite the usefulness of these technologies in policing and law enforcement, their outcomes are not necessarily aligned with the main principles of ethical AI systems, which causes some damage to a group of people (Lunhol & Torhalo, 2024).

In the context of the current stage of AI governance in the Gulf Cooperation Council (GCC) countries, they are in between the early and intermediate governance stages. These nations have national strategies and policy documents; however, most of them rely on soft regulation, referring to strategies, ethical principles, and guidelines rather than detailed and binding AI laws. Additionally, the implementation of AI is uneven across them; some countries, like the UAE and Saudi Arabia, are much more advanced because of

better institutional and political support, investment, and involvement of stakeholders. Qatar has some practical implementation in digital government, and Oman, Bahrain, and Kuwait have positive strategies but less implementation (Albous et al., 2025).

The Sultanate of Oman is an emerging AI adopter within the GCC and, with its ambitious Vision 2040, aims to modernize public services and strengthen national development across multiple sectors (*Oman Vision 2040 Implementation Follow-up Unit*, n.d.). According to UNESCO's Oman AI Readiness Assessment country profile (*Oman | Global AI Ethics and Governance Observatory*, n.d.), Oman has established a strong foundation for AI through its national AI policy, data-protection law, cybersecurity capacity, digital government services, and well-developed internet and data infrastructure. However, the practical implementation of AI governance remains at an early stage, particularly in areas such as enforcement, monitoring, accountability, public participation, AI-specific legal responsibility, research output, commercialization, and AI ethics training. In the law-enforcement context, the clearest publicly reported uses of AI by the Royal Oman Police are currently concentrated in traffic monitoring, road-safety management, and intelligent transport systems (Tridwip, 2025).

Promotion of training and public awareness is significant to have an ethical and legal governance of AI in law enforcement, resulting in a deeper understanding of AI implications (Brand, 2022; Zuwanda et al., 2024). It is also critical to address the current gap existing among law enforcement officials that are lacking the ability to understand how these systems make these decisions and have better control over them, and IT specialists who do not have a suitable knowledge of key ethical principles of AI, like transparency, accountability, fairness, and others, resulting in building these smart systems in total abstraction that makes them to have black-box operation (Lu et al., 2022; Sorell, 2024). Considering the perspective of citizens of various nations, to have a trustworthy and ethical AI system in the law enforcement of a country, a high educational level in Artificial Intelligence is required, enabling the law enforcement officials to safeguard and protect citizens from potential negative effects of AI, since being aware of how these systems operate and make decisions is critical (Ezzeddine et al., n.d.).

Given the established importance of AI ethics education and training in law enforcement, this paper first assesses the current level of responsible AI awareness through a scenario-based survey and semi-structured interviews in the Omani context. Grounded in these empirical findings, it then proposes a structured Ethical AI Awareness and Training Framework (EAATF). Moreover, to address the identified awareness gaps and improve ethical decision-making competencies among law enforcement personnel, this paper aims to propose a Data-Driven Ethical AI Awareness and Training Model (DDEATM) as a personalized and scenario-based learning intervention set in transnational crime contexts.

To accomplish the discussed objectives, this paper is structured as follows: Section 2 states the paper's objectives. Section 3 presents the research questions. Section 4 provides a detailed background to learn more about the key principles and concepts relevant to the study. Section 5 presents the literature review. The research methodology is outlined in Section 6. Results and discussions are mentioned in Sections 7 and 8, respectively. Section 9 presents the conclusion. Finally, Sections 10 and 11 detail the Acknowledgment and References, respectively.

II. Objectives

- To review relevant literature on AI ethics in law enforcement, responsible AI governance, public trust, and ethics-based training models.
- To conduct a scenario-based survey to evaluate the current level of responsible AI awareness among law/security personnel, IT specialists, and general users in the context of law enforcement in the Sultanate of Oman.
- To explore current practices, training approaches, institutional readiness, and practical challenges related to AI ethics and responsible AI use in Omani law enforcement through semi-structured interviews with professionals working in security-relevant roles.

- To propose the Ethical AI Awareness and Training Framework (EAATF) as structured guidelines for the ethical use of AI in law enforcement training, grounded in empirical findings from the Omani context.
- To propose a Data-Driven Ethical AI Awareness and Training Model (DDEATM) as a personalized, scenario-based learning intervention that addresses identified awareness gaps and improves ethical decision-making competencies among law enforcement personnel.

III. Research Questions

1. What do existing studies reveal about AI ethics challenges, responsible AI governance, and ethics-based training models in law enforcement, and to what extent has scenario-based responsible AI training been examined in Omani and wider GCC settings?
2. What is the current level of responsible AI awareness among law/security personnel, IT specialists, and general users in Oman across the domains of data privacy, algorithmic bias, accountability, and critical evaluation of AI outputs?
3. What descriptive changes in responsible AI awareness are observed across the four domains identified in RQ2 following engagement with scenario-based training modules set in transnational crime contexts?
4. What current AI practices, training provisions, accountability mechanisms, ethical concerns, and training-format preferences are reported by professionals working in security-relevant roles in Oman?
5. How can the empirical findings from RQ2 and RQ4 be translated into a structured awareness framework (EAATF) and a personalized, scenario-based training model (DDEATM) appropriate to the differing responsibilities of law/security personnel, IT specialists, and general users?

IV. Background

This section introduces the key concepts and terminology that provide the theoretical background for the study.

A. AI In Law Enforcement

Integrating Artificial Intelligence (AI) elements and applications within legal processes, data analysis, and enhancing security measures to improve efficiency in criminal investigation and activities and enhance judicial capabilities without possessing free will or legal personality, while considering compliance with legal standards, refers to AI in law enforcement (Baltrūnienė, 2023; Chucha, 2023).

Employing AI in law enforcement accelerates criminal investigations, automates vehicle plate number and facial recognition, and helps in crime prevention using predictive policing to anticipate crime types or possible crime locations (Walczak, 2021; Webb et al., 2024). A real-world example of utilizing AI in law enforcement can be mentioned as the TRACE project (Turksen & Kafteranis, n.d.), which helped the Law Enforcement Authorities and Financial Intelligence Units identify, track, and document illicit money flows, particularly in money laundering cases. There are other applications of AI in the context of law enforcement, such as criminal recognition in real time using videos captured from surveillance cameras, finding missing individuals, as well as monitoring and controlling traffic (Lunhol & Torhalo, 2024; Rani, 2024).

B. AI-Assisted Vs. Automated Decision-Making

In AI-Assisted Decision Making (AIADM), humans are the final decision makers, and AI technology is employed as a tool to provide recommendations. This aids team performance by utilizing human judgment and reducing reconciliation expenses. In contrast, Automated Decision Making is where AI independently makes all decisions without any human intervention or judgment (Li et al., 2023; Noti & Chen, 2023; Wolczynski et al., 2023).

Due to the sophisticated, dynamic, and highly context-dependent nature of law enforcement, there is a need for an investigator's knowledge and expertise in the final decision-making. Even with the growth of technology and AI systems in the future, this need remains critical (Nowack et al., 2025). Thus, AI-Assisted Decision Making is more appropriate

than automated decision-making, which allows law enforcement officials to make the final judgment, and AI just facilitates the process of decision-making. Despite the capabilities of AI, human officers are still needed to interpret difficult information, validate AI-generated outputs and identify biases and errors in the system.

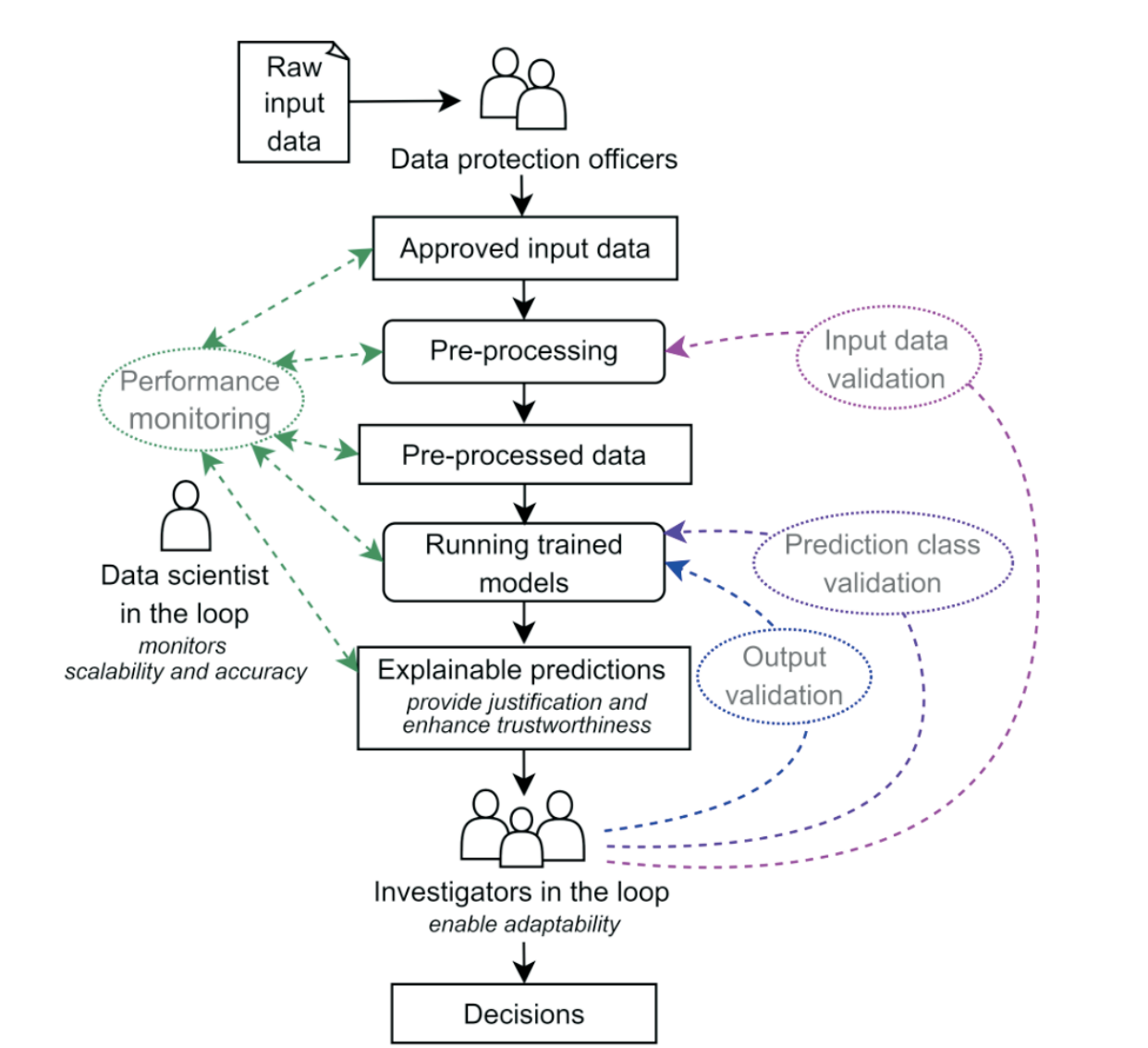


Figure 1. Human-in-the-loop AI-assisted decision-making system for law enforcement. Adapted from **(Nowack et al., 2025)**.

Figure 1 depicts the importance of the human role in the loop of decision-making for validating input data, reviewing AI outputs, monitoring system performance, and making final decisions.

C. Responsible and Trustworthy AI

A group of principles, practices, and governance processes that help the design, development, deployment, and use of AI systems in an ethical, secure, accountable, and trustworthy manner is known as Responsible AI. This type of AI is not only concerned with its accurate operation, but it also attempts to consider the societal values of how it affects people and society, and ensures that the system design respects existing laws and ethical values (*What Is Responsible AI?*, 2024). Table 1 shows some of these critical principles with their definitions that are mainly synthesized from papers (Lu et al., 2022, 2023; Sanderson et al., 2022; Zhang, 2025).

Table 1. Core Responsible AI principles

Principle	Explanation
Fairness & non-discrimination	All AI systems ought to treat different groups of people equally and fairly and should not create or reproduce discriminatory results.
Privacy and data protection	Privacy rights should be respected by AI systems. They must ensure that personal data is collected, processed, stored, and shared lawfully and securely to avoid confidentiality breaches.
Transparency	The functionality of AI systems must be clear and transparent to enable individuals understand how decisions are made by these systems. Transparency replaces the black box nature of AI with visibility.
Explainability	AI systems must be able to explain and articulate how they reached a particular conclusion. The system should provide meaningful reasons for its results and identify the main factors influencing them.
Accountability	The companies, individuals, or organizations that design, purchase, deploy, operate, and monitor must be identifiable and able to provide answers for their system outcomes. In the context of law enforcement, especially, the accountability cannot be fully transferred to AI systems and their algorithms.
Human oversight and control	There should be a human judgment in final decision-making, and AI outputs should be reviewed, rejected, or stopped by these decisions if it is necessary.
Contestability and redress	AI-generated outputs should not be treated as unquestionable facts, and individuals who are highly affected by an AI-supported decision should have an accessible and timely way to question the decision.
Reliability, safety, and security	The consistent performance of AI systems must be guaranteed according to their designed purpose, and they should not create unacceptable risks. They must be accurate, error-free, and robust, and tested before use and continuously monitored after deployment.
Human-centered values and societal wellbeing	Human-centered values, such as human rights, dignity, autonomy, diversity, and cultural values, must be respected by AI systems. These systems must prioritize beneficially to individuals over efficiency.

D. AI as a Socio-Technical System

In the context of law enforcement, AI raises not only algorithmic and technical concerns but also its outputs may be influenced by developers' assumptions, policing practices, legal rules, organizational priorities, human judgment, and the wider social context in which the system operates (Goldberg, 2022; Ugwudike, 2022). For instance, most of the AI training datasets and benchmarks are derived from Western-centric data, which results in cultural biases, underrepresentation of non-Western languages and demographics, and the imposition of Western values onto global users (Zhang, 2025). This is an important matter in law enforcement, since differences in culture, language, behavior, policing practices, and legal systems might affect the interpretation of how suspicious behavior or criminal activity is interpreted.

Hence, an awareness and training framework for AI in law enforcement ought to go beyond preventing bias in datasets; it must also assess the assumptions built into the system, its transparency, institutional objectives, effects on different communities, and the mechanisms available to prevent, correct, and assign responsibility for harmful results.

E. AI Ethics Awareness and Ethical Decision-Making Competency

AI ethics awareness means understanding the implications of using AI in law enforcement to ensure accountability, transparency and protection of human rights. The reason it is important is because of mitigating risks that are related to AI, like privacy violations and biased decision-making (Masters in Criminal Justice, College of Arts and Sciences, Lamar University, Beaumont, Texas, USA et al., 2024). Thus, there is a difference between AI ethics awareness and general knowledge about AI. To ensure there is a development of responsible AI in law enforcement and to appropriately navigate ethical challenges in AI, awareness of ethical AI is crucial (Khan et al., 2022).

Existence of a clear and structured awareness framework enhances the ethical decision-making competency, because these trainings enable the decision makers to learn more about past mistakes made by these AI systems and, through this deeper comprehension, they can make more consistent ethical decisions (De Cremer & Narayanan, 2023).

F. Scenario-based and Data-driven Ethics Training

Scenario-based ethics training can improve decision-making, especially in the context of law enforcement, because it fosters moral evaluation and enhances officer wellness; as a result, the implicit biases will be addressed (Dempsey et al., 2023). This special type of training increases situational awareness and allows officers to develop critical thinking skills useful for communication and ethical decision-making in crisis (Hintze, n.d.; Lavoie et al., 2022).

A data-driven training approach attempts to evaluate the current knowledge level of participants using various methods, including surveys, responses, scenario-based assessments, and interviews, to determine the content and level of training required. This type of training has some positive impacts, such as personalized learning, enhanced learner engagement, demonstrable training effectiveness, and continuous improvements. However, it might have some challenges, especially in sensitive contexts like law enforcement, such as data privacy issues, data security, and change management, since it requires a cultural shift within the organization (*Data-Driven Approach to Training | Grow Learn Connect*, n.d.).

Despite the usefulness of the data-driven training approach, implementing this training method in some areas, like law enforcement, requires some critical considerations to avoid security problems and data privacy issues.

G. Transnational Crime and Cross-border AI Use

A group of activities that span across national borders and more than one country and are related to illicit activities is referred to as transnational crime. Involvement of multiple governments and criminal justice systems is required in this kind of crime, making the process of investigation and prosecution much more complicated (*What Is Transnational Crime?*, n.d.). As previously mentioned, the TRACE project is a great example of how AI-assisted tools can support transnational investigation by analyzing and visualizing complex illicit money flows and patterns across multiple jurisdictions (Turksen & Kafteranis, n.d.).

The use of AI-supported systems in transnational investigations raises a set of problems that do not arise in purely domestic policing. Four are particularly relevant to the Omani context.

The first concerns the lawful movement of data across jurisdictions. In the Sultanate of Oman, the Personal Data Protection Law issued under Royal Decree No. 6/2022 permits a controller to transfer personal data outside the borders of the Sultanate in accordance with the controls and procedures provided for in its Executive Regulations, and prohibits transfer where the data has been processed in violation of the Law or where transfer is likely to cause harm to the data subject (Sultanate of Oman, 2022). The Executive Regulations, issued under Ministerial Decision No. 34/2024, provide that the consent of the data owner is sufficient for transfer outside Oman, provided that the transfer does not prejudice national security or the higher interests of the country (Ministry of Transport, Communications

and Information Technology [MTCIT], 2024). Where an AI-supported system requires data held in another jurisdiction, or produces outputs derived from such data, these controls determine whether the exchange may lawfully occur (MTCIT, n.d.).

The second concerns the legal basis for cooperation between authorities. Oman is a party to the Arab Convention on Combating Information Technology Offences, ratified by Royal Decree No. 5/2015, under which a party may request another to investigate, access, seize, secure, or disclose information technology data located within the territory of the requested party (Sultanate of Oman, 2015; Dominioni, 2023). Formal mutual legal assistance of this kind operates on timescales that may not correspond to the operational tempo at which AI-generated alerts are produced, creating a practical gap between what a system indicates and what an officer may lawfully act upon.

The third concerns evidential admissibility. An output generated by an AI system in one jurisdiction, using training data and validation standards determined there, may be introduced in proceedings in another. Where the basis of the output cannot be explained to the standard required by the receiving forum, its evidential value is uncertain. This difficulty compounds the explainability problems already identified in domestic policing contexts (Herrewijnen et al., 2024).

The fourth concerns divergent governance regimes. AI systems used in cross-border investigations may be developed under one regulatory framework and deployed under another, with different requirements concerning human oversight, transparency, and permissible data use. In the absence of a harmonized regional framework, officers may be required to act on outputs produced under standards that differ from those applying in their own jurisdiction (Albous et al., 2024).

These problems share a common characteristic: each requires the officer to make a judgment that domestic training does not prepare them to make. This provides the rationale for treating transnational contexts as a distinct training requirement rather than as a variant of domestic practice, and it informs the design of the cross-border accountability scenario within DDEATM.

V. Systematic Literature Review

This section will provide an overview of related papers to this study by highlighting their findings and results and comparing them with other research papers.

A. Inclusion and Exclusion Criteria

This paper conducted a systematic literature review that included various papers published between 2018 and 2026, including empirical studies, reviews, and conference papers, and peer-reviewed articles related to responsible AI in law enforcement and transactional crime. The review also considered relevant projects and frameworks proposed or implemented in different countries to provide a broader understanding of the current state of responsible AI in law enforcement from both global and Sultanate of Oman perspectives. This enabled the study to examine the connection between theoretical concepts of responsible AI and their practical application in law enforcement contexts. Non-English, irrelevant to the research focus, non-peer-reviewed, and inaccessible in full-text papers were excluded from this review.

B. Search Strategy

The literature search was conducted across multiple academic databases and search engines, such as IEEE Xplore, SpringerLink, ScienceDirect, Google Scholar, ResearchGate, BASE (Bielefeld Academic Search Engine), SciSpace, and Semantic Scholar. Various combinations of keywords were utilized related to AI, responsible AI, AI ethics, law enforcement, policing, criminal justice, transactional crime, AI governance, privacy, bias, transparency, accountability and human rights. Initially, simple search strings employing Boolean Operators (AND, OR, NOT) were used to find related papers (e.g., "responsible AI" AND "law enforcement") , also the filtering features available in some search-engines like Google Scholar was used to filter papers published between 2018 and 2026 to meet mentioned Inclusion criteria. In addition, the reference lists of identified relevant studies were examined to identify further publications related to the research topic.

C. Study Selection Process

As shown in the PRISMA diagram Figure 1, 74 studies were initially identified from various sources. After removing 26 irrelevant, inaccessible, and duplicate records, 48 articles proceeded to the screening stage. 6 more records were excluded in screening by reviewing titles and abstracts, leaving 42 research papers for full-text assessment. At the eligibility stage, 43 papers were evaluated for inclusion, and 16 of them were excluded because they were used solely as background or contextual sources and were therefore not eligible for inclusion in the systematic literature review, leaving 26 studies for the final review.

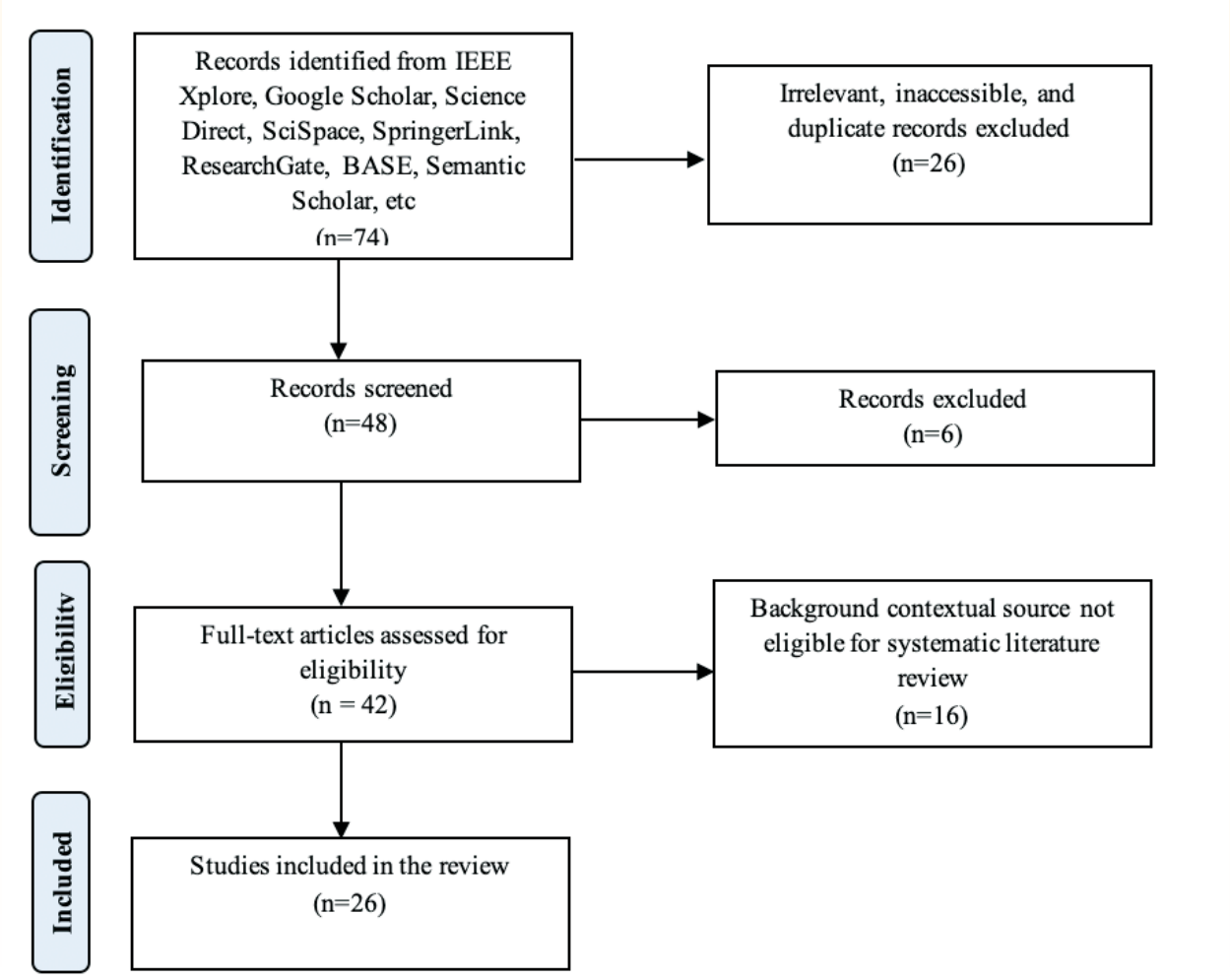


Figure 2. PRISMA flow diagram

D. Review Findings and Synthesis

Previous research has emphasized both opportunities and risks related to the use of AI in policing (Webb et al., 2024). As mentioned, AI technology can improve the speed, accuracy, and efficiency of law enforcement activities using applications such as predictive policing, automated image analysis, facial recognition, and traffic monitoring. For instance, Automated Plate Recognition (ANPR) is capable of identifying vehicle plate numbers in real time, while AI-supported traffic monitoring systems have been used in Delhi, India (Rani, 2024). Automated Facial Recognition technologies can also assist police in identifying individuals of interest and locating missing persons or children (Rani, 2024; Sorell, 2024). Despite these advantages, AI systems may reproduce bias that is contained in the historical datasets, resulting in unfair classification of some people and the over-policing of marginalised communities. An appropriate example could

be the Harm Assessment Risk Tool (HART), which was designed to assist police officers in evaluating a person's risk of reoffending and identifying eligibility for a crime-diversion programme. However, they stopped using it in 2021, since it was disproportionately classifying some groups as being at high risk (Webb et al., 2024). This paper (Ugwudike, 2022) adds more to the risks caused by these systems by arguing that unfair outcomes are not produced only by historically biased datasets; they may also result from assumptions and theories embedded in the system's design by developers and creators of these AI systems.

Training plays a pivotal role in having an ethical AI system in law enforcement, since it promotes awareness among the public, law enforcement officials, and IT specialists by highlighting the AI implications and helps to build trust and increase the engagement of society (Brand, 2022; Zuwanda et al., 2024). This paper (Sorell, 2024) examined how the UK police projects are ethically reviewed by using the ALGOCARE checklist that contains questions analyzing whether an AI system is safe, fair, legal, and appropriate before using it. The paper studied two ethics commities including the West Midlands Police Data Ethics Committee and the Biometrics and Forensics Ethics Group. It identified that there is a knowledge gap between police officers who may not fully understand data science, AI limitations, or data protection, and data science and IT specialists who may not fully comprehend policing procedures, legal limits, and police ethics.

Adding to the paper by (Sorell, 2024), which used ALGOCARE practically, this article (Goldberg, 2022) proposed two approaches for evaluating the ethics of AI systems in policing, including top-down and bottom-up approaches. The top-down approach starts with established ethical principles and frameworks, like ALGOCARE, and attempts to apply them to a specific ethical issue that arise from that situation. On the other hand, the bottom-up approach begins with identifying ethical issues in a real policing case. Explainability and co-design with police users are also emphasized in this paper to make sure that AI tools are appropriate for operational work. The case-based assessment method proposed in this article supports the need for practical and scenario-based ethics training that teaches police officers and technical specialists how to recognize ethical risks and evaluate AI systems within a real policing context.

The crucial gap between high-level ethical AI principles and their practical implementation in real systems was highlighted by this article (Sanderson et al., 2022). Although the responsible AI principles (e.g., fairness, transparency, accountability, etc) are widely recognized, there is a lack of concrete technical translation of these concepts into operational practices. This issue was also reflected in the discussion of (Selbst et al., 2019), which explains that engineering approaches attempt to translate and simplify complex social and ethical problems into mathematical and measurable technical variables, resulting in the implementation of these systems in total abstraction. In areas like law enforcement, this abstraction can create framing and formalism problems. To address the mentioned gap, this paper (Lu et al., 2022) creates a software engineering roadmap for responsible AI. After carefully reviewing 159 primary studies to identify the existing responsible AI solutions, the authors proposed a roadmap containing three main perspectives: governance, development process, and system design. The suggested roadmap indicates that responsible AI must go beyond theoretical concepts and be applied across the full AI lifecycle through ethical requirements, testing, human oversight, monitoring, and accountability. Role-specific ethics training for managers, developers, data scientists, and operators was recommended as well by this paper, which shows the importance of training and awareness. This approach was developed further by(Lu et al., 2023) through reusable practices such as ethical risk assessment, stakeholder engagement, documentation, explainability, testing, human oversight, and continuous monitoring.

Based on the reviewed studies, the literature emphasized that the preparation of ethical concepts ought to occur before AI systems are implemented in risky fields like law enforcement. This study (Rani, 2024) argues that all applications of AI in policing must be supported by a suitable education level, human supervision, strong legal safeguards, and a diverse dataset to reduce the risk of bias and human rights violations. Similarly, this article (Lu et al., 2023) explains that employees need enough and appropriate ethical risk awareness to make responsible choices during the development and use of AI systems.

The context of some advanced countries, like the US (Agbadamasi et al., 2025), shows that there is a need for a dedicated responsible AI framework, ethical safeguards, and professional awareness before deployment, as the existing laws are not sufficient to govern modern AI systems. The development and application of Explainable AI in law enforcement make this need especially clear. Thanks to this article (Herrewijnen et al., 2024) that found that police officers prefer useful and understandable explanations over highly technical graphs, scores or individual words based on interviews conducted with them using an AI-supported document-filtering system. The results demonstrate that while officers must continue to be in charge of crucial policing choices, explainability tools should be developed in accordance with officers' operational demands, expertise, and domain knowledge. Because inadequate comprehension of AI outputs may lead to wrong interpretation, overreliance, bias, and broader social harm, the study argues for the necessity of practical training in advance of deployment. Overall, these studies show that AI ethics awareness should not stay theoretical but instead provide developers, technical experts, law enforcement officers, and organizational decision-makers with role-specific and useful advice.

The citizens' perspectives in 8 different countries were examined by this paper (Ezzeddine et al., n.d.), which conducted approximately 111 interviews to understand the viewpoints of people toward data collection and the use of AI by law enforcement agencies. One of the main concerns of participants was the lack of AI knowledge among law enforcement officials to protect citizens from the potential negative effects of AI. They mentioned that there should be a suitable training procedure to enable officers and personnel of law enforcement who are not coming from an IT background to understand how decisions are made in these systems. Out of all safeguards suggested by interviews conducted by the authors of the paper, education was the most suggested one.

Due to this study (Zhang, 2025), which aimed to analyze major AI ethics challenges and make a comparison between global regulatory approaches across different countries using policy analysis and case studies, aiming to identify the gap between ethical principles and real enforcement and finally provide some suggestions for more trustworthy AI governance. The findings of this paper emphasized that despite the widespread theories about ethical AI and its principles, such as fairness, transparency, accountability, privacy, and human oversight, they are not appropriately enforced and implemented in practice. Additionally, differences between regulatory approaches in various countries were highlighted. Some regions like the EU follow a stronger precautionary approach, while the US heavily relies on voluntary and market-based regulation, and China uses centralized mandatory control. Moreover, the paper supports the importance of awareness by mentioning that there is a need for interdisciplinary cooperation among developers, policymakers, legal experts, ethicists, civil society groups, and the public to have an ethical and responsible AI.

Two neural network models were developed in this study (Walczak, 2021) that used crime data from Detroit. One of these models was concentrating on predicting the type of crime from the time and location of the crime, while the other model was anticipating the possible location of the crime based on the type and time. Although some of the models were able to provide some useful information to police, their accuracy was limited. Therefore, the author suggested that these systems must be used as tools only for assisting police officers, not independently making final decisions. In contrast to (Walczak, 2021) project, which was mainly concentrated on the technical side, the TRACE project proposed by (Turksen & Kafteranis, n.d.) has brought together police authorities, technical developers, legal experts, ethicists, and social scientists to work closely with each other to develop an AI tool for investigating money laundering and illicit money flows. TRACE project indicated that technical performance alone is not enough to have a responsible and ethical AI, especially in fields like law enforcement.

Thanks to this paper (Albous et al., 2025), which studied how six GCC countries, including Oman, the UAE, Saudi Arabia, Qatar, Bahrain, and Kuwait, are managing and regulating AI. National AI strategies and government policies published among 2018 and 2024 were examined by this paper. The findings of the paper indicate that a soft regulation approach is mostly

employed in these nations; in other words, they are dependent on strategies, ethical guidelines, and general principles instead of strict and binding AI laws. The author highlighted that most of the GCC strategies emphasize the importance of ethical ideas (e.g., privacy, fairness, transparency, accountability, human oversight, and safety), but when it comes to practice, they are not clearly implemented and enforced.

Based on the existing literature, it is clear that AI can improve law enforcement, especially in countries such as the Sultanate of Oman, where the use of these systems is still developing. However, if AI is not implemented appropriately, it may create risks related to bias, privacy, transparency, accountability, and human rights. The literature review also highlighted practical knowledge gaps among police officers and IT specialists, as well as difficulties in applying responsible AI principles in real systems. However, limited research has examined these issues in Oman or developed a locally adapted, scenario-based training approach. This study addresses this gap through the proposed EAATF and DDEATM.

VI. Methodology

A. *Research Design*

This study utilized a mixed-methods research design consisting of a one-group pretest and post-test approach. Quantitative data were collected from the participants to measure changes in participants' AI ethics awareness after completing the scenario-based training, while qualitative data were gathered to explore ethical concerns, professional experiences, and training needs. The quantitative and qualitative findings were then triangulated to provide a more comprehensive understanding of AI ethics awareness in the Omani context.

B. *Participants and Sampling*

The study involved 40 survey participants from three background groups: IT/Computer Science, law/security, and other fields representing general users. Participants were recruited using non-probability convenience sampling based on their accessibility and relevance to the study. Additionally, three professionals with experience in law enforcement, security, or public service in Oman were purposively selected for the semi-structured interviews.

C. *Survey Instrument*

The primary data collection tool employed in this study is a structured electronic questionnaire developed specifically for this research and administered via Google Forms in both Arabic and English to ensure accessibility for the target population. The instrument was organized into five sequential sections, including participant information, pre-evaluation, scenario-based Training, post-evaluation, and an open question. Table 2 illustrates the Structure of the Survey Instrument.

Table 2. Structure of Survey Instrument

Section	Component	Content and Purpose
1	Participant Information	Collected participants' fields of study or work, IT/Computer Science, Law/Security, or Other, and their prior exposure to AI systems.
2	Pre-Test	Included four multiple-choice questions measuring baseline awareness of data privacy, algorithmic bias, accountability, and critical evaluation of AI outputs. Each question contained three options, with one representing ethically sound practice.
3	Scenario-Based Training	Presented three transnational-crime scenarios. Surveillance Bias: responding to a facial-recognition match with 70% accuracy. Data Privacy: deciding whether private user data could be used in a cybercrime investigation. Cross-Border Accountability: responding to an AI system identifying a suspect in another country without sufficient evidence.
4	Post-Test	Repeated the four pre-test questions to allow direct comparison of participants' awareness before and after the scenario-based training.
5	Open-Ended Question	Invited participants to identify the most significant ethical challenges associated with AI use in law enforcement, providing qualitative data to complement the pre/post results.

D. *Semi-Structured Interviews*

Semi-structured interviews were conducted to explore current AI practices, ethical awareness, and training needs in greater depth. The bilingual interview guide contained 14 questions covering participants' roles, AI systems in use, transnational crime applications, existing training, organizational policies, accountability, ethical challenges, and preferred training design. Interviews lasted between 15 and 20 minutes and were conducted face-to-face or through voice calls, according to participant preference.

E. *Data Analysis*

A mixed-methods approach to data analysis was used in this study. For every survey item, Google Forms' built-in features produced response counts and percentages. The Matplotlib and pandas libraries in Python were utilized to create figures and analyze participant attributes. To find changes after the scenario-based training, descriptive comparisons of the pre-test and post-test outcomes were made. Braun and Clarke's (2006) six-phase theme analysis was used to inductively analyze open-ended survey responses and interview data. In order to determine persistent awareness gaps and training needs, the quantitative and qualitative results were finally triangulated.

F. *Ethical Considerations*

This study was conducted in full accordance with the ethical guidelines and approval procedures of the Research and Studies Center at Sultan Qaboos Police Sciences Academy. Informed consent was obtained from all survey participants through an explanatory statement displayed at the beginning of the electronic questionnaire. For interview participants, verbal informed consent was obtained prior to the commencement of each interview.

No personal identifying information was collected at any stage of the data collection process. All survey responses were anonymous, and interview participants were assigned codes (P2, P3, P1) to ensure anonymity in data recording, analysis, and reporting. The study involved no vulnerable populations, no deception, no physical or psychological risk to participants, and no collection of sensitive personal data. Participation was entirely voluntary, with no material incentives offered.

G. *Framework and Model Development*

The EAATF and DDEATM were developed through a structured five-stage process. First, survey and interview findings were analysed to identify gaps in data privacy awareness, accountability understanding, and structured ethics training. Second, these gaps were mapped against established international AI ethics frameworks (*Artificial Intelligence Toolkit*, n.d.; *Ethics Guidelines for Trustworthy AI | Shaping Europe's Digital Future*, n.d.; Oswald et al., 2018). Third, recognized learning and feedback theories (*Bloom et al -Taxonomy of Educational Objectives.Pdf*, n.d.; Hattie & Timperley, 2007; Oswald et al., 2018)were used to organize the six principles of the Ethical AI Awareness and Training Framework (EAATF). Fourth, these principles were operationalized through the seven components of the Data-Driven Ethical AI Awareness and Training Model (DDEATM). Finally, the model was personalized for law-enforcement officers, IT specialists, and general users according to their different roles and training needs.

Stage	Development activity	Outcome	Supporting references
Gap identification	Analysed survey and interview findings	Identified privacy, accountability, and training gaps	Present study
Ethical mapping	Compared the findings with established ethical guidance	Selected relevant ethical principles	(*Artificial Intelligence Toolkit*, n.d.; *Ethics Guidelines for Trustworthy AI | Shaping Europe's Digital Future*, n.d.; Oswald et al., 2018)
Pedagogical structuring	Applied established learning and feedback theories	Developed the six EAATF principles	(*Bloom et al -Taxonomy of Educational Objectives. Pdf*, n.d.; Hattie & Timperley, 2007; Kolb, 2014)
Model operationalization	Converted the principles into training activities	Developed the seven DDEATM components	Present study
Personalization	Considered different professional roles and needs	Created three personalized user tracks	Present study

VII. Results

This section presents the findings of the study across two strands: quantitative results from the scenario-based survey instrument and qualitative findings from the semi-structured interviews and open-ended survey question.

A. Participant Characteristics

A total of 40 participants completed the survey. Law/ Security personnel represented the largest group (n = 15, 37.5%), followed by participants from other fields (n = 14, 35.0%) and IT/Computer Science specialists (n = 11, 27.5%). This distribution captured different professional perspectives on AI ethics awareness in Oman, as illustrated in Figure 2.

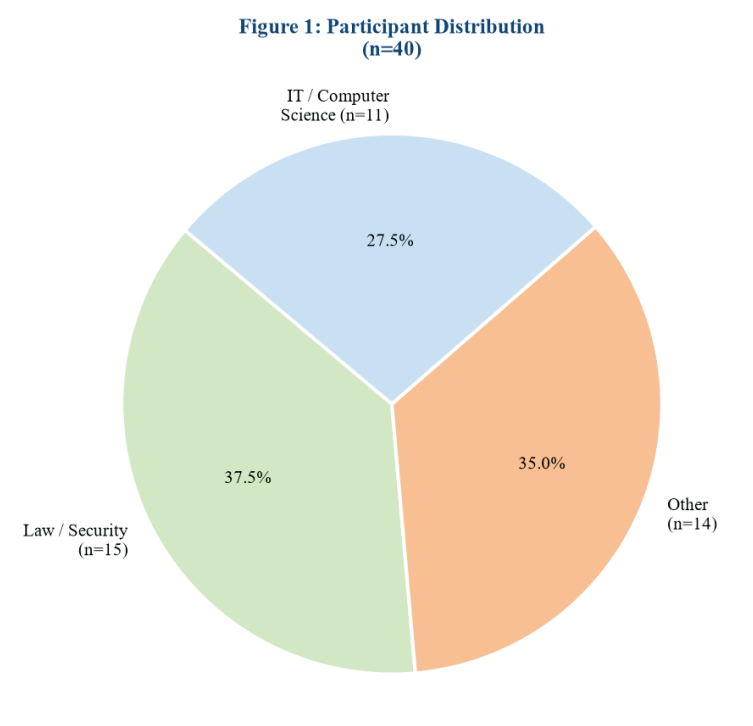


Figure 3. Distribution of Participants by Professional Background (n=40)

Most participants reported prior knowledge of AI (n = 35, 87.5%), while nearly all had previously interacted with AI systems (n = 38, 95.0%). These findings indicate that the sample possessed foundational AI familiarity, providing an appropriate basis for assessing ethical awareness. Figure 3 presents participants' prior AI knowledge and interaction.

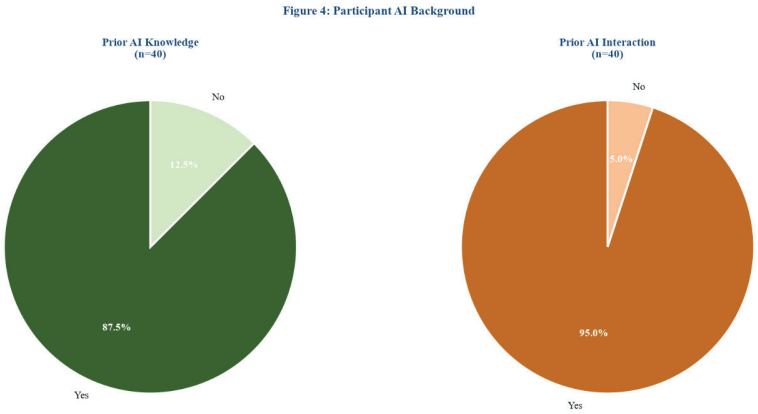


Figure 4. Participants' Prior AI Knowledge and Interaction with AI Systems (n=40)

B. Pre-Evaluation Results

Pre-test scores revealed varying levels of AI ethics awareness across the four assessed domains prior to engagement with the scenario-based training modules. The weakest area of awareness was data privacy, with only 37.5% of participants selecting the ethically correct response (Q1), identifying that the use of personal data without consent for crime detection is not ethically justified. Accountability was correctly identified by 80.0% of participants (Q3), while algorithmic bias (Q2) and critical evaluation of AI outputs (Q4) demonstrated the highest baseline awareness levels at 92.5% and 95.0%, respectively. The aggregate pre-test mean score was 3.05 out of 4 (76.3%). Table 3 summarizes the pre-evaluation results by item.

Table 3. Pre-Evaluation Results by Item (n=40)

Item	Ethical Domain	Correct Responses	Percentage
Q1	Data Privacy	15/40	37.5%
Q2	Algorithmic Bias	37/40	92.5%
Q3	Accountability	32/40	80.0%
Q4	Critical Evaluation of AI	38/40	95.0%
Mean		**30.5/40**	**76.3%**

C. Scenario-Based Training Results

Participants demonstrated strong ethical decision-making across all three scenario-based training modules. As illustrated in Figure 4, Scenario 1 (Surveillance Bias) recorded the highest correct-response rate at 95.0%, indicating that most participants recognized the need to verify facial-recognition results with additional evidence. Scenario 3 (Cross-Border Accountability) followed at 90.0%, showing awareness of the need to coordinate with competent authorities before acting on insufficient AI-generated evidence. Scenario 2 (Data Privacy) recorded an 87.5% correct-response rate, demonstrating participants' understanding that only legally authorized data should be used in cybercrime investigations.

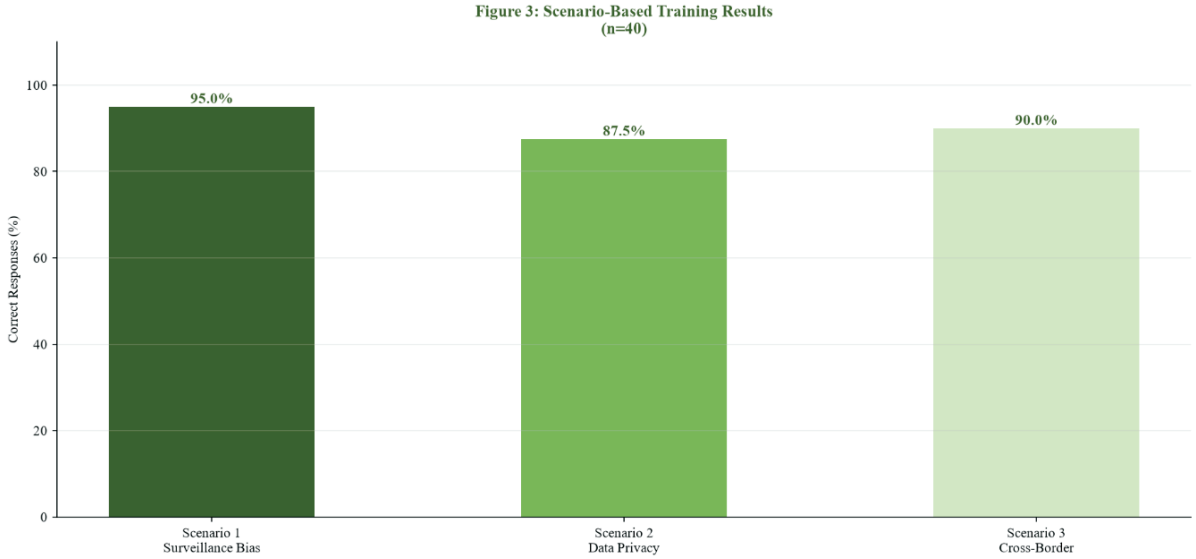


Figure 5. Percentage of participants selecting the ethically appropriate response in each scenario-based training module (n = 40).

D. Post-Evaluation Results and Pre/Post Comparison

Following engagement with the three scenario-based training modules, post-test scores demonstrated improvement across three of the four assessed domains. Data privacy awareness showed the most notable increase, rising from 37.5% to 47.5% (+10.0 percentage points), representing the largest single-domain improvement and reflecting the effectiveness of the data privacy scenario in raising awareness of this critical ethical principle. Accountability improved from 80.0% to 87.5% (+7.5 percentage points). Algorithmic bias awareness increased marginally from 92.5% to 95.0% (+2.5 percentage points), likely reflecting a ceiling effect given the already high baseline score. Critical evaluation of AI outputs showed a slight decrease from 95.0% to 92.5% (-2.5 percentage points), which may similarly reflect ceiling effects and minor response variability in a domain where baseline awareness was already near-maximal. The aggregate mean score increased from 3.05 out of 4 (76.3%) to 3.23 out of 4 (80.6%), representing an improvement of 0.18 points, or 4.3 percentage points. Figure 5 illustrates the pre/post comparison across the four assessed domains.

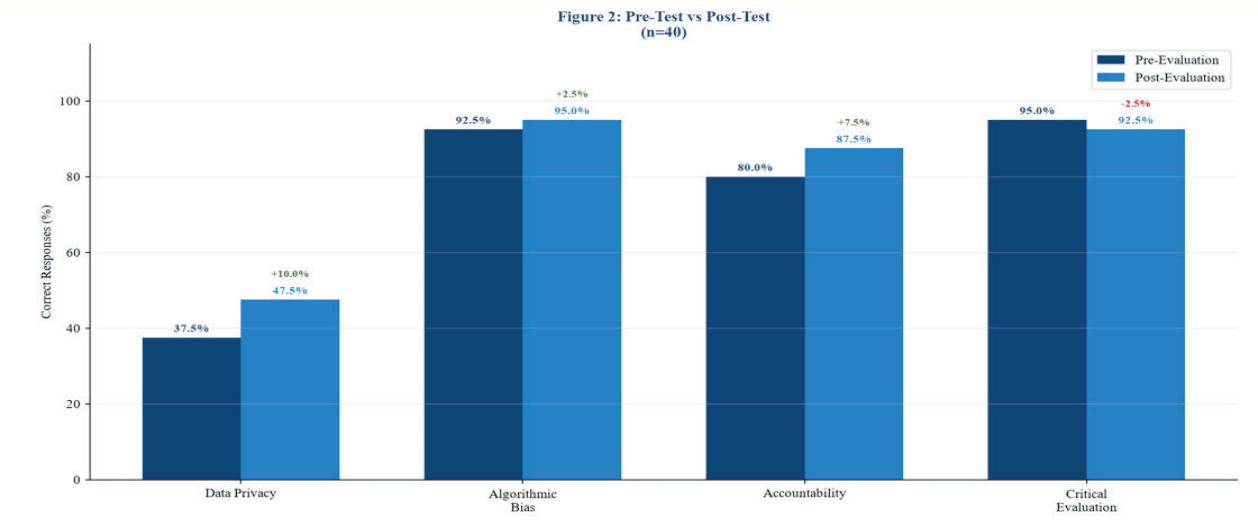


Figure 6. Pre-evaluation and post-evaluation AI ethics awareness across the four assessed domains (n = 40).

E. Qualitative Findings from Open-Ended Survey Responses

Thematic analysis of participants' responses to the open-ended survey question revealed five recurring themes relating to the ethical challenges of AI use in law enforcement. The most frequently cited theme was privacy and surveillance concerns, with participants expressing concern about the potential for AI-enabled mass surveillance and unauthorized data collection. Algorithmic bias and fairness constituted the second theme, with participants noting the risk of discriminatory outcomes arising from biased training data. Transparency and accountability emerged as the third theme, with participants highlighting the opacity of AI decision-making processes and the difficulty of assigning responsibility for AI-assisted decisions that lead to negative outcomes. Over-reliance on technology was identified as the fourth theme, with participants emphasizing the risk of officers deferring excessively to AI outputs at the expense of human judgment and professional discretion. The fifth theme, absence of legal and regulatory frameworks, reflected participants' concern about the inadequacy of existing laws and policies to govern AI use in law enforcement in the Omani context.

F. Qualitative Findings from Semi-Structured Interviews

Semi-structured interviews were conducted with three participants (P1, P2, and P3) representing professional backgrounds in healthcare, aviation, and administration/security, respectively. The thematic analysis identified six key themes, summarized in Table 4.

Table 4. Themes Identified from the Semi-Structured Interviews

Theme	Main finding	Supporting evidence
Limited AI deployment	AI adoption in the participants' professional environments remained limited or at an early stage.	P2 reported no AI use in transnational crime cases, P1 reported no daily AI use, and P3 acknowledged AI-related systems but could not provide details because of confidentiality.
Absence of structured AI ethics training	Participants had not received dedicated training addressing responsible and ethical AI use.	P1 reported no AI ethics training, while P2 and P3 described existing training as general internet-security guidance and operational warnings.
Human oversight as a shared value	All participants believed that AI outputs should be questioned and verified by humans.	P1 and P2 stated that AI results should be accepted only after verification, while P3 confirmed that officers could question AI-supported results.
Ambiguity of accountability	Participants could not identify a clear party or mechanism responsible for harmful AI-supported decisions.	None of the three participants provided a specific accountability mechanism.
Multidimensional ethical challenges	Ethical concerns were viewed as interconnected rather than isolated.	Participants identified limited training, unclear policies, privacy concerns, and over-reliance on technology. P3 considered all the listed challenges relevant.
Preference for practical training	Participants preferred practical and application-oriented training approaches.	Suggested formats included on-the-job training, online modules, real policing scenarios, case studies, workshops, simulations, and continuous follow-up.

Overall, participants demonstrated high familiarity with AI and strong baseline awareness of algorithmic bias and critical evaluation, but considerably lower awareness of data privacy. Participants generally selected ethically appropriate responses across the three scenarios, while the aggregate mean increased from 76.3% to 80.6% following the intervention. The qualitative findings identified privacy, bias, transparency, accountability, over-reliance on technology, and limited regulation as major concerns. The interviews further revealed limited structured AI ethics training, unclear accountability mechanisms, strong support for human oversight, and a preference for practical, scenario-based training.

VIII. Discussions

This section interprets the findings in relation to the research objectives and existing literature, discusses the implications of the proposed EAATF and DDEATM, and considers the broader significance of the study for responsible AI governance in Omani law enforcement.

A. AI Ethics Awareness Among Study Participants

The quantitative findings reveal a mixed but informative picture of AI ethics awareness among the participating law/security personnel, IT specialists, and general users. The aggregate pre-evaluation mean of 76.3% indicates relatively high overall performance across the four assessed ethical domains. However, the considerable variation between data privacy at 37.5% and critical evaluation of AI outputs at 95.0% shows that participants' awareness was uneven and that some ethical areas were less clearly understood than others.

The data privacy item produced the lowest baseline result, with only 37.5% of participants selecting the ethically appropriate response. This is important because privacy and lawful data use are recognized as fundamental requirements for trustworthy AI and responsible AI use in law enforcement (*Artificial Intelligence Toolkit*, n.d.; *Ethics Guidelines for Trustworthy AI | Shaping Europe's Digital Future*, n.d.). The result suggests that privacy should receive

greater attention before AI-supported systems become more widely adopted in law-enforcement activities.

The high baseline results for algorithmic bias at 92.5% and critical evaluation of AI outputs at 95.0% were encouraging. These findings may partly reflect participants' existing familiarity with AI, as 87.5% reported prior AI knowledge and 95.0% had interacted with AI systems. However, recognizing an ethical principle does not necessarily demonstrate the ability to apply it in operational situations. Previous examination of algorithmic risk-assessment systems in policing has demonstrated the importance of considering accuracy, proportionality, human judgment, and operational context when using AI-supported outputs (Oswald et al., 2018). This distinction supports using realistic scenarios that require participants to apply ethical principles rather than simply recognize their definitions.

B. Preliminary Outcomes of Scenario-Based Training

The descriptive pre/post results indicate preliminary improvement following participation in the three scenario-based modules. The aggregate mean increased by 4.3 percentage points, from 76.3% to 80.6%. The largest observed improvement occurred in data privacy, which increased by 10.0 percentage points from its comparatively low baseline. This suggests that targeted exposure to realistic privacy dilemmas may help address areas in which awareness is initially weaker.

These findings are consistent with the theoretical foundations of DDEATM. Experiential learning theory indicates that active engagement with realistic experiences can support learning more effectively than the passive presentation of abstract information (Kolb, 2014). The scenario-based approach used in this study and formalized within DDEATM requires participants to consider realistic transnational crime situations and select an appropriate course of action. DDEATM further develops this approach by proposing immediate and contextualized feedback, which is recognized as an important influence on learning (Hattie & Timperley, 2007). However, the specific contribution of feedback should be examined through future implementation and evaluation.

Participants selected ethically appropriate responses at high rates across all three scenarios: 95.0% for surveillance bias, 90.0% for cross-border accountability, and 87.5% for data privacy. These results indicate that participants were generally able to apply ethical reasoning when presented with contextualized law-enforcement dilemmas. Nevertheless, the high results may partly reflect the relatively straightforward response options. Future scenarios should therefore incorporate more complex situations involving competing legal, operational, and ethical considerations.

Improvement was not consistent across all four domains. Critical evaluation decreased by 2.5 percentage points, while algorithmic bias increased by only 2.5 percentage points. These limited changes may reflect ceiling effects because both domains already had high baseline results, as well as minor response variability. Therefore, the observed changes should be interpreted as preliminary descriptive findings rather than conclusive evidence of training effectiveness.

C. Qualitative Insights and Triangulation

The qualitative findings from the open-ended responses and semi-structured interviews complemented the quantitative results. Survey participants identified privacy, surveillance, algorithmic bias, transparency, accountability, over-reliance on technology, and limited legal frameworks as important ethical concerns. These themes closely correspond to the ethical requirements emphasized in international trustworthy and responsible AI guidance (*Artificial Intelligence Toolkit*, n.d.; *Ethics Guidelines for Trustworthy AI | Shaping Europe's Digital Future*, n.d.)and in previous research on algorithmic risk assessment in policing (Oswald et al., 2018).

The interviews provided additional contextual insight. All three interviewees reported an absence of dedicated, structured AI ethics training. P1 reported receiving no training related to AI ethics or the limitations of digital systems, while P2 and

P3 described existing training as general internet-security guidance and operational warnings rather than specific instruction on responsible AI principles. These findings provide contextual support for the awareness gaps observed in the survey and demonstrate the difference between receiving general technology training and developing practical AI ethics competency.

Agreement among all three interview participants on the importance of human oversight was another important finding. Participants consistently stated that AI outputs should be verified before being accepted or used to support consequential decisions. This position aligns with the human-oversight requirements expressed in international responsible AI guidelines (*Artificial Intelligence Toolkit*, n.d.; *Ethics Guidelines for Trustworthy AI | Shaping Europe's Digital Future*, n.d.). It also provides a valuable foundation for incorporating human verification and final human decision-making into EAATF and DDEATM.

The interviews also revealed ambiguity regarding accountability. None of the three participants could identify a clear party or mechanism responsible for a harmful AI-supported decision. Although this finding cannot represent the entire Omani law-enforcement sector, it reflects the importance of accountability and clearly defined human responsibility emphasized in existing ethical frameworks (*Artificial Intelligence Toolkit*, n.d.; *Ethics Guidelines for Trustworthy AI | Shaping Europe's Digital Future*, n.d.). It supports the inclusion of accountability content within EAATF and confirms the relevance of the cross-border accountability scenario within DDEATM.

Interview responses were recorded in Arabic and translated by the authors. Beyond the absence of structured training, P1 reported that no established procedures existed for ensuring responsible use of digital systems in their working environment, and that no guidance was provided regarding erroneous or biased outputs. P2 characterized available training as addressing the safe use of the internet and smart systems, while P3 described it as consisting of operational warnings. These accounts indicate that existing provision addresses general digital safety rather than responsible AI use as a distinct competency, supporting the Evaluate principle of EAATF, which establishes a participant's baseline awareness prior to instruction rather than presupposing that prior information-technology training has addressed ethical content.

Although all participants supported human verification, their responses indicate that such verification is understood as conditional rather than categorical. P1 and P2 both described accepting AI outputs only after confirming the accuracy of the information, with P2 adding that continued follow-up was necessary. P3 indicated that outputs may be questioned because such systems remain human-supported, while maintaining that they should nonetheless be used. P2 articulated the underlying tension directly, noting that the system reduces workload, that trust requires time to develop, and that most tasks demand ongoing monitoring. When asked what an AI ethics training programme should prioritize, P1 responded that officers should not depend on such systems entirely. This suggests that the practicability of verification determines whether it occurs, a consideration consistent with the finding of Herrewijnen et al. (2024) that officers' engagement with AI-supported outputs is shaped by operational demands rather than by technical explanation alone. The observation supports the retention of explicit verification steps throughout the DDEATM scenarios and the Reflect principle of EAATF.

Both P1 and P3 indicated that disclosure was not permitted in response to questions concerning the application of digital systems to cross-border cases, with P3 extending this constraint to the systems currently in use and to procedures for addressing erroneous outputs. This is understood as a characteristic of the research environment rather than a deficiency in the data. It indicates that operational AI use within Omani law enforcement is governed by confidentiality requirements that limit what externally conducted research may establish, and it reinforces the case for awareness and training initiatives developed and delivered within the institution.

D. Implications for Transnational Contexts

The accountability divergence reported in Section VII-C acquires additional significance in cross-border settings. Where participants operating within a single institutional environment attributed responsibility variously to the officer, the information-technology function, and to no identifiable party, the introduction of a second jurisdiction adds further candidates: the requesting authority, the authority operating the system, and the entity that supplied the training data. An accountability model that is unsettled domestically is unlikely to resolve when responsibility is distributed across national boundaries, a requirement emphasized in established responsible AI guidance (Artificial Intelligence Toolkit, n.d.; Ethics Guidelines for Trustworthy AI | Shaping Europe's Digital Future, n.d.; Oswald et al., 2018).

The disclosure constraints reported by P1 and P3 are similarly consequential. Where participants were unable to describe whether digital systems are applied to cross-border cases, the same confidentiality requirements would govern any attempt to reconstruct the decision chain following a harmful outcome in a transnational investigation. This reinforces the argument for embedding accountability content within institutional training rather than relying on external review.

DDEATM addresses these conditions through its cross-border accountability scenario, in which participants respond to an AI system identifying a suspect in another jurisdiction on insufficient evidence. Future development of the model should extend this scenario to incorporate the data-transfer controls and cooperation mechanisms described in Section III-G, so that participants engage with the legal basis for cross-border action rather than with the ethical dimension alone.

E. Implications for EAATF and DDEATM

The integrated findings provide a preliminary empirical basis for EAATF and DDEATM. The comparatively low baseline result for data privacy confirms the relevance of the privacy scenario and indicates that privacy should receive greater emphasis in future versions of the training. Similarly, uncertainty surrounding responsibility for harmful AI-assisted decisions supports including accountability content and further developing the cross-border accountability scenario.

The absence of dedicated AI ethics training among the interview participants also supports the sequential structure of EAATF. The framework begins by evaluating participants' existing awareness, develops foundational knowledge, and allows participants to apply that knowledge through realistic situations. This structure is informed by established principles of knowledge assessment and progressive learning (*Bloom et al -Taxonomy of Educational Objectives*, n.d.) and experiential learning (Kolb, 2014). Its Feedback and Reflect principles encourage participants to understand the consequences of their choices (Hattie & Timperley, 2007), while the Update principle allows training content to respond to technological, legal, and operational developments.

Interview participants' preferences for on-the-job training, simulations, realistic scenarios, workshops, case studies, and online modules support the practical design of DDEATM. These preferences indicate that the proposed format corresponds with the learning needs expressed by participants. Its personalized tracks for law-enforcement officers, IT specialists, and general users also recognize that these groups have different responsibilities and levels of technical and operational knowledge. However, the effectiveness of these tracks must be evaluated through future implementation with larger and more representative samples.

F. Relevance to Oman and the GCC Context

The findings have potential relevance to Oman and similar contexts across the GCC where responsible AI practices and professional training continue to develop. The privacy, accountability, and training gaps identified in this study suggest that transforming high-level ethical principles into practical competency requires targeted and context-sensitive educational interventions. Aligning EAATF and DDEATM with established international guidance (*Artificial*

Intelligence Toolkit, n.d.; *Ethics Guidelines for Trustworthy AI | Shaping Europe's Digital Future*, n.d.) also provides a foundation for adapting the proposed approach to different legal, institutional, and professional environments.

The proposed framework and model may contribute to national objectives concerning innovation, institutional capacity building, and responsible governance. By providing a structured pathway for developing AI ethics awareness among law-enforcement personnel, IT specialists, and general users, EAATF and DDEATM may support a more human-centred, accountable, and ethically prepared security environment. Further testing and institutional validation will be required before wider adoption, but this study provides an initial foundation for responsible AI training in the Omani context.

IX. Conclusion

This study used semi-structured interviews with three experts and a multilingual survey with 40 participants to investigate ethical AI awareness and training needs in the context of law enforcement in Oman. Participants did well in identifying algorithmic bias and critically assessing AI outputs, but data privacy was the weakest area (37.5%), according to the results, which demonstrated inconsistent ethical awareness. Following the scenario-based exercises, this outcome rose by 10 percentage points. Limited structured training, unclear accountability, a preference for hands-on learning, and agreement among respondents that humans should validate AI outputs and maintain ultimate decision-making authority were all highlighted by the qualitative findings.

Based on the discussed findings, this paper proposed an Ethical AI Awareness and Training Framework (EAATF) and a Data-Driven Ethical AI Awareness and Training Model (DDEATM). EAATF provides six training principles, namely: Evaluate, Awareness, Apply, Feedback, Reflect, and Update, while DDEATM translates them into a seven-component, personalized training intervention for law-enforcement personnel, IT specialists, and general users.

Due to the limited survey and interview sample sizes, the use of only three scenarios, and the lack of a control group, the results should be regarded with caution. It's also possible that relevant studies published in regional languages were not considered due to the restricted evaluation of non-English literature. Larger and more varied samples, more scenarios, and literature in different languages should also be included in future research. Subject to institutional clearance and suitable ethical, security, and confidentiality standards, it should also test the full EAATF and DDEATM with the Royal Oman Police. In addition to raising awareness among IT professionals and the public, such implementation could promote ethical AI training for law enforcement personnel.

X. Acknowledgment

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